

Explore File and Data Encryption

Cybersecurity Investigation and Encryption Analysis Report

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1. Executive Summary

This project focused on the investigation and analysis of encrypted file handling and transfer operations within a simulated enterprise network environment using Cisco Packet Tracer and Linux-based cryptographic tools.

The exercise demonstrated how confidential information can be encrypted before transmission, uploaded to a remote FTP server, downloaded by authorized users, and later decrypted using OpenSSL.

The project also highlighted the security limitations of legacy protocols such as FTP and examined the importance of secure encryption, password-based authentication, and confidentiality protection in cybersecurity operations.

Practical experience was gained in:

- Cryptographic operations
- Secure file handling
- Encryption and decryption workflows
- Linux command-line operations
- OpenSSL usage
- FTP authentication
- Confidential data protection
- Security analysis

2. Project Objectives

The objectives of this project were:

Part 1 – Discover FTP Credentials for Mary

- Access encrypted FTP login information
- Decrypt the credentials using OpenSSL
- Authenticate to the FTP server

Part 2 – Upload Confidential Data Using FTP

- Verify encrypted file contents
- Upload encrypted data to the FTP server
- Analyze FTP transmission security

Part 3 – Discover FTP Credentials for Bob

- Decrypt Bob's FTP login information
- Authenticate to the server securely

Part 4 – Download Confidential Data Using FTP

- Download encrypted files from the server
- Validate successful transfer operations

Part 5 – Decrypt Sensitive Data

- Obtain decryption keys
- Decrypt confidential files using OpenSSL
- Recover readable plaintext information

3. Background Information

Organizations frequently transmit confidential information across networks. To protect this information from unauthorized access, encryption technologies are used to convert readable plaintext into unreadable ciphertext.

In this Packet Tracer activity, encrypted customer information was transferred between two employees, Mary and Bob, using FTP and OpenSSL.

Although the files themselves were encrypted, the activity also demonstrated that FTP remains an insecure protocol because usernames, passwords, and commands can still be transmitted in clear text.

The lab provided practical exposure to:

- Data confidentiality
- Password-based encryption
- Secure file handling
- Encryption weaknesses
- Secure authentication procedures

4. Tools and Technologies Used

Tool / Technology

Purpose

Cisco Packet Tracer	Network simulation environment
OpenSSL	Encryption and decryption operations
Linux Terminal	Command-line cryptographic tasks
FTP	File transfer operations
AES-256-CBC	Encryption algorithm
PBKDF2	Password-based key derivation
Base64 Encoding	Encoded ciphertext formatting
VirtualBox	Virtual machine environment
CSE-LABVM	Linux cybersecurity virtual machine

5. Network Environment Overview

The simulated network environment consisted of:

- Branch Office workstations
- FTP Server
- Encrypted confidential files
- User authentication systems
- Remote file transfer operations

Two users were involved in the process:

- Mary – uploader of confidential data
- Bob – authorized recipient of confidential data

The primary file transferred during the exercise was *clientinfo.enc*

This file contained encrypted customer information.

6. Part 1 – Discovering Mary's FTP Credentials

Objective

To decrypt Mary's encrypted FTP login credentials in order to access the FTP server.

Procedure

Mary's encrypted credentials were stored inside the file *maryftplogin.txt*

The encrypted text was copied into the Linux virtual machine and decrypted using OpenSSL.

OpenSSL Command Used

```
echo 'ENCRYPTED_TEXT' | openssl aes-256-cbc -pbkdf2 -a -d
```

Decryption Password

maryftp123

Analysis

The encryption utilized:

- AES-256-CBC encryption
- PBKDF2 password-based key derivation
- Base64 encoding

This process demonstrated secure credential storage and password-based cryptography.

Security Importance

Encrypted credential storage prevents attackers from immediately reading sensitive authentication data if the file is exposed.

7. Part 2 – Uploading Confidential Data Using FTP

Objective

To upload an encrypted confidential file to the FTP server.

Procedure

The encrypted file *clientinfo.enc* was verified before transmission.

The FTP connection was established using *ftp 10.0.3.30*

Mary's decrypted credentials were used for authentication.

The encrypted file was uploaded using the FTP **put** command.

Security Observation

The confidential file itself was encrypted; however, FTP traffic still exposed usernames, passwords, FTP commands as well as Session metadata in plaintext.

Analysis

If a threat actor intercepted the FTP session, they would likely observe authentication traffic, file names, commands as well as network metadata

This demonstrates why FTP is considered insecure in modern enterprise environments.

Command Prompt

```
C:\>ftp 10.0.3.30
Trying to connect...10.0.3.30
Connected to 10.0.3.30
220- Welcome to PT Ftp server
Username:mary
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 10.0.3.30:
0      : asa842-k8.bin                      5571584
1      : asa923-k8.bin                      30468096
2      : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
3      : c1841-ipbase-mz.123-14.T7.bin       13832032
4      : c1841-ipbasek9-mz.124-12.bin       16599160
5      : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6      : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7      : c2600-i-mz.122-28.bin              5571584
8      : c2600-ipbasek9-mz.124-8.bin        13169700
9      : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10     : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11     : c2800nm-ipbase-mz.123-14.T7.bin    5571584
12     : c2800nm-ipbasek9-mz.124-8.bin      15522644
13     : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14     : c2950-i6q4l2-mz.121-22.EA4.bin     3058048
15     : c2950-i6q4l2-mz.121-22.EA8.bin     3117390
16     : c2960-lanbase-mz.122-25.FX.bin     4414921
17     : c2960-lanbase-mz.122-25.SEE1.bin   4670455
18     : c2960-lanbasek9-mz.150-2.SE4.bin   4670455
19     : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20     : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21     : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22     : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23     : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24     : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25     : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26     : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
27     : ir800-universalk9-mz.SPA.155-3.M   61750062
28     : ir800-universalk9-mz.SPA.156-3.M   63753767
29     : ir800_yocto-1.7.2.tar             2877440
```

- FTP authentication

Command Prompt

```

29 : ir800_yocto-1.7.2.tar                      2877440
30 : ir800_yocto-1.7.2_python-2.7.3.tar         6912000
31 : pt1000-i-mz.122-28.bin                      5571584
32 : pt3000-i6q4l2-mz.121-22.EA4.bin            3117390
ftp>put clientinfo.enc

Writing file clientinfo.enc to 10.0.3.30:
File transfer in progress...

[Transfer complete - 564 bytes]

564 bytes copied in 0.119 secs (4739 bytes/sec)
ftp>dir

Listing /ftp directory from 10.0.3.30:
0 : asa842-k8.bin                               5571584
1 : asa923-k8.bin                               30468096
2 : c1841-advipservicesk9-mz.124-15.T1.bin      33591768
3 : c1841-ipbase-mz.123-14.T7.bin               13832032
4 : c1841-ipbasek9-mz.124-12.bin                16599160
5 : c1900-universalk9-mz.SPA.155-3.M4a.bin      33591768
6 : c2600-advipservicesk9-mz.124-15.T1.bin      33591768
7 : c2600-i-mz.122-28.bin                      5571584
8 : c2600-ipbasek9-mz.124-8.bin                 13169700
9 : c2800nm-advipservicesk9-mz.124-15.T1.bin    50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin    33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin            5571584
12 : c2800nm-ipbasek9-mz.124-8.bin              15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin      33591768
14 : c2950-i6q4l2-mz.121-22.EA4.bin            3058048
15 : c2950-i6q4l2-mz.121-22.EA8.bin            3117390
16 : c2960-lanbase-mz.122-25.FX.bin             4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin           4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin           4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin    8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin     10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin        33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin       83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin     505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG        159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG        184530138
26 : clientinfo.enc                             564
27 : ir800-universalk9-bundle.SPA.156-3.M.bin   160968869

```

- FTP upload operation & File listing verification

8. Part 3 – Discovering Bob’s FTP Credentials

Objective

To decrypt Bob’s encrypted FTP login credentials.

Procedure

Bob’s encrypted credentials were stored in *bobftplogin.txt*

The encrypted data was decrypted using OpenSSL.

OpenSSL Command Used

```
echo 'ENCRYPTED_TEXT' | openssl aes-256-cbc -pbkdf2 -a -d
```

Decryption Password

bobftp123

Analysis

This phase reinforced the importance of:

- Secure password handling
- Credential confidentiality
- Password-based encryption

It also demonstrated how encrypted credentials can safely be distributed and stored.

9. Part 4 – Downloading Confidential Data Using FTP

Objective

To download the encrypted confidential file from the FTP server.

Procedure

Bob authenticated to the FTP server using his decrypted credentials.

The encrypted file *clientinfo.enc* was downloaded using the FTP get command.

Security Analysis

Although the file contents remained encrypted, the FTP protocol still transmitted:

- User credentials
- Session commands
- Transfer metadata

without encryption.

Cybersecurity Implication

Attackers monitoring network traffic could:

- Capture usernames and passwords
- Observe transferred files
- Replay authentication attempts
- Conduct credential theft attacks

10. Part 5 – Decrypting Sensitive Files

Objective

To decrypt the confidential customer information file.

Procedure

Bob received the decryption key via email.

The encrypted file contents were copied into the Linux virtual machine.

The following OpenSSL command was used:

```
openssl aes-256-cbc -pbkdf2 -a -d -in clientinfo.enc -out clientinfo.txt
```

The decryption password provided by Mary was entered.

A new plaintext file was successfully generated *clientinfo.txt*

Analysis

This demonstrated:

- Symmetric encryption workflows
- Secure information recovery
- Password-based decryption operations
- Confidential data handling

Cryptographic Technologies Used

AES-256-CBC

Advanced Encryption Standard using:

- 256-bit encryption keys
- Cipher Block Chaining mode

PBKDF2

Password-Based Key Derivation Function 2:

- Strengthens password-derived encryption keys
- Increases resistance to brute-force attacks

Base64 Encoding

Used to safely represent encrypted binary data in text format.

11. Security Weaknesses Identified

FTP is Insecure

FTP does not encrypt authentication credentials, session commands and metadata

This exposes organizations to residential theft, packet sniffing, session hijacking and Man-in-the-middle attacks.

Integrity Validation Not Guaranteed

The encryption process used in the activity does not fully validate file integrity. Attackers may potentially alter encrypted files without immediate detection.

Email-Based Key Sharing

Sending encryption keys through email introduces risks such as Email interception, Credential leakage and Unauthorized access

12. Security Recommendations

Replace FTP with Secure Protocols

Use SFTP, FTPS or SCP

These protocols encrypt credentials, commands and file transfers

Implement Multi-Factor Authentication

MFA reduces the risk of unauthorized access even if credentials are compromised.

Use Authenticated Encryption

Adopt modern authenticated encryption methods such as AES-GCM and ChaCha20-Poly1305. These provide both confidentiality and integrity verification

Deploy Centralized Logging

Organizations should monitor Authentication attempts, File transfer activity as well as Suspicious login behavior using SIEM platforms.

Restrict User Permissions

Apply the Principle of Least Privilege to minimize unnecessary access.

Use Secure Key Management

Encryption keys should never be shared through insecure communication channels.

13. Lessons Learned

This project demonstrated several important cybersecurity concepts:

- Encryption protects confidentiality but not necessarily integrity
- FTP is vulnerable to interception attacks
- OpenSSL provides powerful cryptographic functionality

- Password-based encryption improves security
- Secure file transfer protocols are critical in enterprise environments

The project also strengthened practical Linux and cybersecurity analysis skills.

14. Skills Demonstrated

Technical Skills

- Cryptography
- OpenSSL Operations
- Linux Administration
- FTP Operations
- Secure File Handling
- Encryption and Decryption
- Command-Line Analysis
- Cybersecurity Documentation

Cybersecurity Skills

- Confidentiality Protection
- Authentication Analysis
- Security Assessment
- Threat Awareness
- Network Security Fundamentals
- Data Protection

15. Conclusion

This project provided practical experience in encryption, file transfer security, and cryptographic operations within a simulated enterprise environment.

The activity successfully demonstrated how encrypted data can be securely transferred between authorized users while also highlighting the security weaknesses of legacy protocols such as FTP.

By completing this project, practical understanding was gained in:

- OpenSSL encryption workflows
- Password-based cryptography

- Secure data handling
- File confidentiality protection
- Security weaknesses in insecure protocols

The project further strengthened foundational skills required for cybersecurity operations, SOC analysis, digital forensics, and incident response.